

LAB # 04

EXERCISE:

Consider Triangle problem : /* Design and develop a program in a language of your choice to solve the triangle problem defined as follows : Accept three integers which are supposed to be the three sides of triangle and determine if the three values represent an equilateral triangle, isosceles triangle, scalene triangle, or they do not form a triangle at all. Derive test cases for your program based on decision-table approach, execute the test cases and discuss the results

Test Case Name :Equivalence class Analysis for triangle problem Test Data : Enter the 3 Integer Value(a , b And c) Pre-condition : $1 \leq a \leq 10$, $1 \leq b \leq 10$ and $1 \leq c \leq 10$ and $a < b + c$, $b < a + c$ and $c < a + b$

Code:

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int side1, side2, side3;
```

```
    cout << "Enter value of side1: ";
```

```
    cin >> side1;
```

```
    cout << "Enter value of side2: ";
```

```
    cin >> side2;
```

```
    cout << "Enter value of side3: ";
```

```
    cin >> side3;
```

```
    if (side1 >= 1 && side1 <= 10 && side2 >= 1 && side2 <= 10 && side3 >= 1 && side3 <= 10) {
```

```
        if (side1 < side2 + side3 && side2 < side1 + side3 && side3 < side1 + side2) {
```

```

if (side1 == side2 && side2 == side3) {
    cout << "Equilateral Triangle " << endl;
} else if (side1 == side2 || side1 == side3 || side2 == side3) {
    cout << "isosceles Triangle " << endl;
} else {
    cout << "scalene Triangle " << endl;
}
} else {
    cout << "Cannot form a triangle." << endl;
}
} else {
    cout << "Error inavlid input." << endl;
}
return 0;
}

```

Equivalence Class Partitonaing

Test Case Id	Description	a	b	c	Expected Output	Actual Output	Status	Comments
1	Valid Equilateral Triangle	6	6	6	Should display "Equilateral Triangle"	Displayed	Pass	
2	Valid Isosceles Triangle	6	6	8	Should display "Isosceles Triangle"	Displayed	Pass	
3	Valid Scalene Triangle	5	7	9	Should display "Scalene Triangle"	Displayed	Pass	
4	Invalid Triangle (Sum of two sides $\leq$ third side)	3	3	7	Should display "Cannot form a Triangle"	Displayed	Pass	
5	Invalid Input (Negative Side)	-3	5	6	Should display "Error: Invalid Input"	Displayed	Pass	
6	Invalid Input (Positive Side)	15	5	6	Should display "Error: Invalid Input"	Displayed	Pass	

Test Cases

TC-001

```
Enter value of side1: 6
Enter value of side2: 6
Enter value of side3: 6
Equilateral Triangle
```

TC-002

```
Enter value of side1: 6
Enter value of side2: 6
Enter value of side3: 8
isosceles Triangle
```

TC-003

```
Enter value of side1: 5
Enter value of side2: 7
Enter value of side3: 9
scalene Triangle
```

TC-004

```
Enter value of side1: 3
Enter value of side2: 3
Enter value of side3: 7
Cannot form a triangle.
```

TC-005

```
Enter value of side1: -3
Enter value of side2: 5
Enter value of side3: 6
Error inavlid input.
```

TC-006

```
Enter value of side1: 15
Enter value of side2: 5
Enter value of side3: 6
Error inavlid input.
```

Boundary Value Analysis

Test Case Id	Description	a	b	c	Expected Output	Actual Output	Status	Comments
1	Valid Equilateral Triangle	6	6	6	Should display "Equilateral Triangle"	Displayed	Pass	
2	Valid Isosceles Triangle	6	6	8	Should display "Isosceles Triangle"	Displayed	Pass	
3	Valid Scalene Triangle	5	7	9	Should display "Scalene Triangle"	Displayed	Pass	
4	Invalid Triangle (Sum of two sides $\leq$ third side)	3	3	7	Should display "Cannot form a Triangle"	Displayed	Pass	
5	Invalid Input (lower bound-1)	0	5	6	Should display "Error: Invalid Input"	Displayed	Pass	
6	Invalid Input (upper bound+1)	11	5	6	Should display "Error: Invalid Input"	Displayed	Pass	
5	Invalid Input (lower bound)	1	5	6	Should display "Error: Invalid Input"	Displayed	Pass	
6	Invalid Input (upper bound)	10	5	6	Should display "Error: Invalid Input"	Displayed	Pass	

Test Cases

TC-001

```
Enter value of side1: 6
Enter value of side2: 6
Enter value of side3: 6
Equilateral Triangle
```

TC-002

```
Enter value of side1: 6
Enter value of side2: 6
Enter value of side3: 8
isosceles Triangle
```

TC-003

```
Enter value of side1: 5
Enter value of side2: 7
Enter value of side3: 9
scalene Triangle
```

TC-004

```
Enter value of side1: 3
Enter value of side2: 3
Enter value of side3: 7
Cannot form a triangle.
```

TC-005

```
Enter value of side1: 0
Enter value of side2: 5
Enter value of side3: 6
Error inavlid input.
```

TC-006

```
Enter value of side1: 11
Enter value of side2: 5
Enter value of side3: 6
Error inavlid input.
```

TC-007

```
Enter value of side1: 1
Enter value of side2: 5
Enter value of side3: 6
Cannot form a triangle.
```

TC-008

```
Enter value of side1: 10
Enter value of side2: 5
Enter value of side3: 6
scalene Triangle
```

